

Statistics 5810/6810: Graphics 2

Case: East Bay Housing Market

```
load(url("http://www.math.usu.edu/  
~symanzik/teaching/2012_stat5810/  
LectureNotes/SFHousing.rda"))
```

Warning: It's BIG

San Francisco Chronicle listings



Data

- Record: house sold in a particular time period
 - Over 200,000 houses
 - Subset to a dozen cities in the East Bay – about 25,000 houses
- Variables:
- City
 - County
 - Price
 - # bedrooms
 - Lot square footage
 - and 10 more

Relationship between city and sale price

Data types:

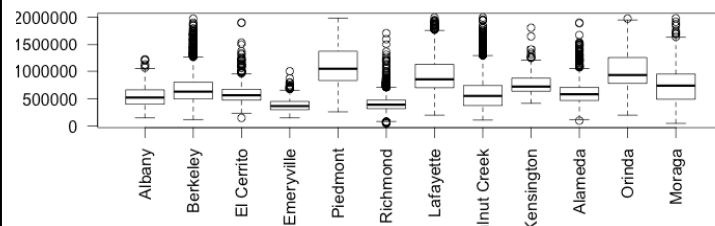
City - factor

Sale price - numeric

Examine a subset of the cities

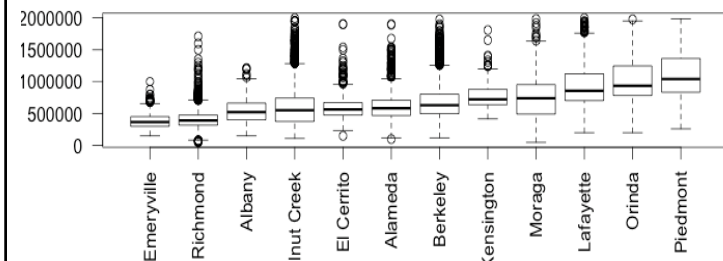
```
someCities = c("Albany", "Berkeley", "El
Cerrito", "Emeryville", "Piedmont",
"Richmond", "Lafayette", "Walnut Creek",
"Kensington", "Alameda", "Orinda", "Moraga")
shousing =
housing[housing$city %in% someCities &
housing$price < 2000000,]
dim(shousing)
[1] 20415 15
```

Boxplots



```
shousing$city = factor(shousing$city, levels
= someCities)
#adjust factors to those in someCities
boxplot(shousing$price ~ shousing$city, las =
2)
```

Cities ordered by median price



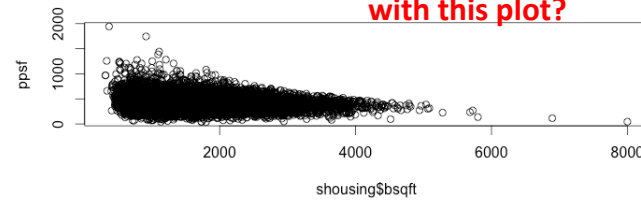
```
citymedian = tapply(shousing$price, shousing$city, median,
na.rm=TRUE)
cityOrder = order(citymedian)
shousing$cityO = factor(shousing$city, levels =
levels(shousing$city)[cityOrder], ordered = TRUE)
boxplot(shousing$price ~ shousing$cityO, las = 2)
```

Relationship between price per square foot and total square foot

Both are quantitative

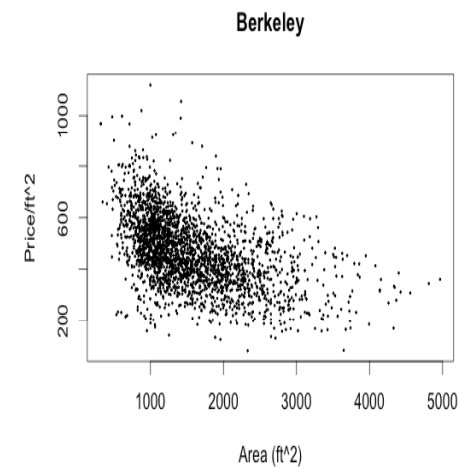
```
ppsf = shousing$price/shousing$bsqft
plot(ppsf ~ shousing$bsqft)
```

**WHAT's Wrong
with this plot?**



Scatter plot

```
plot(ppsf ~ shousing$bsqft, # plot y against
x
pch=19, # change plotting character to solid
circle
cex = 0.2, # shrink plotting character to
20%
subset = shousing$city == "Berkeley",
# Plot a subset of
records
main="Berkeley", # title of plot
xlab="Area (ft^2)", # label for x
axis
```



Relationships between more than 2 variables

- Qualitative information can be conveyed in plots through color, plotting symbol, juxtaposed panels
- The following plot uses information from 4 variables: city, number of bedrooms, lot size (sq ft), and price per square ft

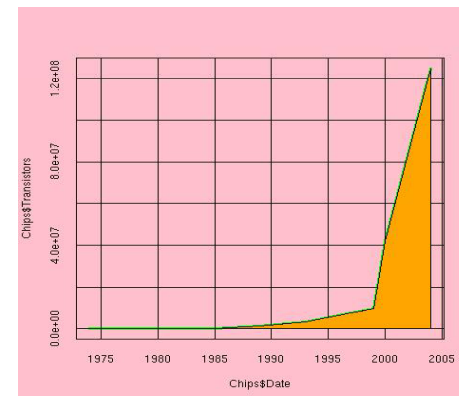
What do you see?



Summary of graph relationships between two variables

- Two Qualitative variables
 - mosaicplot, side-by-side barplots
- One Quantitative and one Qualitative
 - Boxplots, dotcharts, multiple density plots
- Two Quantitative variables
 - Scatter plot, line plot (for temporal data)

What do you think of this plot?



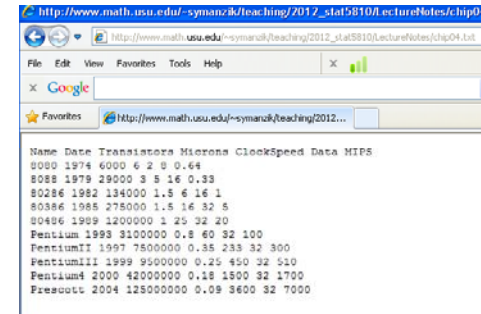
FIND 5 things that you would change

Let's fix it!

Making good plots is an iterative process
Goal is to convey a message as clearly as possible

Visit the website

http://www.math.usu.edu/~symanzik/teaching/2012_stat5810/LectureNotes/chip04.txt



Name	Date	Transistors	Microns	ClockSpeed	Data	MIPS
8080	1974	6000	6	2.8	0.64	
8088	1979	29000	3	5	16	0.33
80286	1982	134000	1.5	6	16	1
80386	1985	275000	1.5	16	32	5
80486	1989	1200000	1	25	32	20
Pentium	1993	3100000	0.8	60	32	100
PentiumII	1997	7500000	0.35	233	32	360
PentiumIII	1999	9500000	0.25	450	32	510
Pentium4	2000	42000000	0.18	1300	32	1700
FreeSoc	2004	125000000	0.09	3600	32	7000

Read it into R

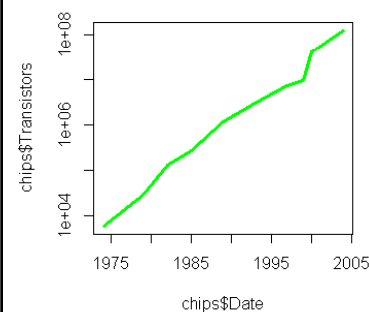
```
chips =
read.table("http://www.math.usu.edu/~symanzik/teaching/2012_stat5810/LectureNotes/chip04.txt", header = TRUE)
```

```
class(chips)
[1] "data.frame"
```

```
names(chips)
[1] "Name"          "Date"
"Transistors"
[4] "Microns"       "ClockSpeed"   "Data"
[7] "MIPS"
```

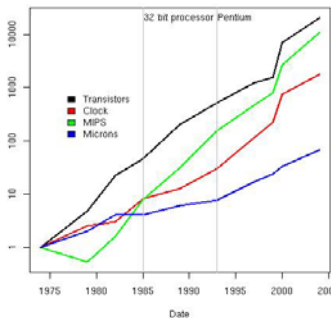
```
dim(chips)
```

This is pretty easy to get



```
plot(chips$Date,
chips$Transistors,
type = "l",
lwd = 3,
col = "green",
log="y")
```

How can we improve it even more?



- Add more data
- Add legend for different information
- Add reference lines for important dates
- Rescale according to reference year (e.g., rescale to 1 or 100 in 1974)

Review Plotting Functions

- `hist()` histogram
- `boxplot()` boxplot
- `dotchart()` dotchart
- `plot()` for scatter plots, line plots, density plots
- `barchart()`
- `pie()`
- `mosaicplot()`
- `abline()` add line to canvas
- `points()` add points to canvas
- `lines()` add line segments to canvas
- `text()` add text to canvas

Review Plot Arguments

`?plot.default`

- `type = "l"` "p" for points, "l" for lines, "n" for nothing
- `ylim = c(0, 1)` the range for the scale of the axis
- `xlab = "x axis label"`
- `main = "plot title"`
- `col = vector of colors`
- `log = "y"` use log scale on y axis, can be
- `lwd = 2` thickness of line
- `pch = 19` plotting character – check other numbers
- `cex = 0.5` character magnification
- `lty = 2` type of line – check other numbers
- Further options via `par()` command – see help page!

Graph Construction

- 3 Properties of good graph construction
 - Data stand out
 - Facilitate comparison
 - Information rich

Choosing the Scale of the Axes

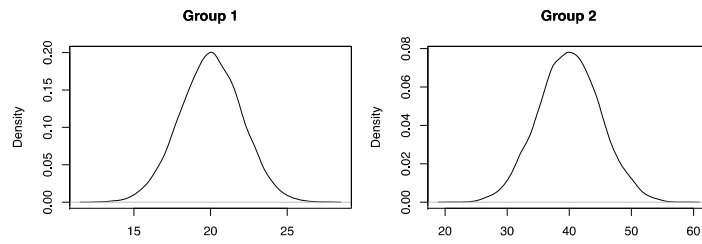
- Include all or nearly all of the data
- Fill data region
- Origin need not be on the scale (except for histograms & bar charts)
- Choose a scale that improves resolution

Eliminate superfluous material

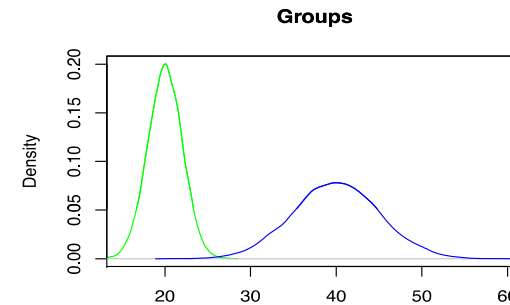
- Chart junk – stuff that adds no meaning, e.g. butterflies on top of barplots, background images
- Extra tick marks and grid lines
- Unnecessary text and arrows
- Decimal places beyond the measurement error or the level of difference

Facilitate Comparisons

Put Juxtaposed plots on same scale



Make it easy to distinguish elements of superposed plots (e.g. color)



Choosing the Scale

- Keep scales on x and y axes the same for both plots to facilitate the comparison
- Zoom in to focus on the region that contains the bulk of the data
- These two principles may go counter to one another
- Keep the scale the same throughout the plot (i.e. don't change it mid-axis)

How to make a plot information rich

- Describe what you see in the **Caption**
- Add context with **Reference Markers** (lines and points) including text
- Add **Legends** and **Labels**
- Use color and plotting symbols to add more information
- Plot the same thing more than once in different ways/scales
- Reduce clutter

Captions

- Captions should be comprehensive
- Self-contained
- Captions should:
 - Describe what has been graphed
 - Draw attention to important features
 - Describe conclusions drawn from graph

Good Plot Making Practice

- Put major conclusions in graphical form
- Provide reference information
- Proof read for clarity and consistency
- Graphing is an iterative process
- Multiplicity is OK, i.e. two plots of the same variable may provide different messages
- Make plots data rich

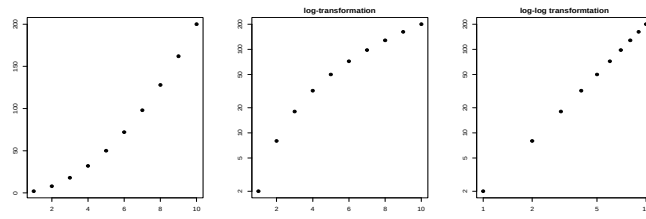
Perception

Color, shape (including banking) can affect **comparisons**

Banking: Aspect Ratio

- The height/width of the data region should be selected to be about 1 so that the trend line is at about 45 degrees.
- The aspect ratio affects our visual decoding of the rate of change
- The banking to 45 degrees helps us see rate of change
- The ability to effectively judge rate of change allows us to see important patterns in data

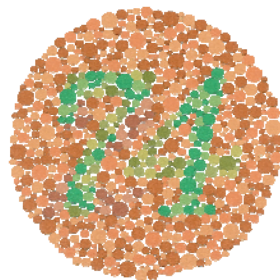
Bank to 45 degrees



Color

Color Guidelines

- Choosing a set of colors which work well together is a challenging task for anyone who does not have an intuitive gift for color
- 7-10% of males are red-green color blind.



Data Type and Color

- Qualitative – Choose a **qualitative** scheme that makes it easy to distinguish between categories
- Quantitative – Choose a color scheme that implies magnitude.
 - Does the data progress from low to high? Use a **sequential** scheme where light colors are for low values
 - Do both low and high value deserve equal emphasis? Use a **diverging** scheme where light colors represent middle values

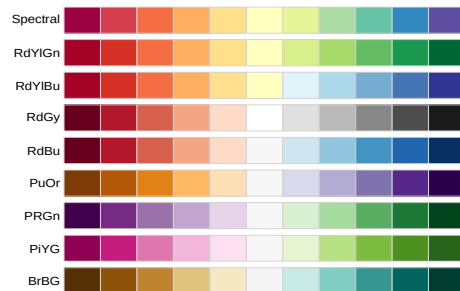
ColorBrewer

- Cindy Brewer at Penn State developed a tool called ColorBrewer
- Web version at <http://colorbrewer2.org/>

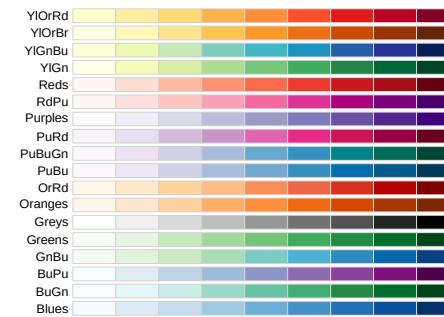
Brewer's Qualitative Palettes



Brewer's Diverging Palettes



Brewer's Sequential Palettes

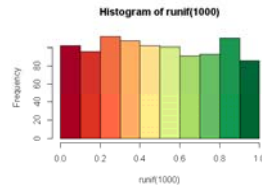


Brief look at how to use the special colors from Brewer's palettes in R

```
library(RColorBrewer)
```

```
mycolors = brewer.pal(10, "RdYlGn")
```

```
hist(runif(1000), breaks = seq(0, 1, by = 0.1),  
     col = mycolors)
```

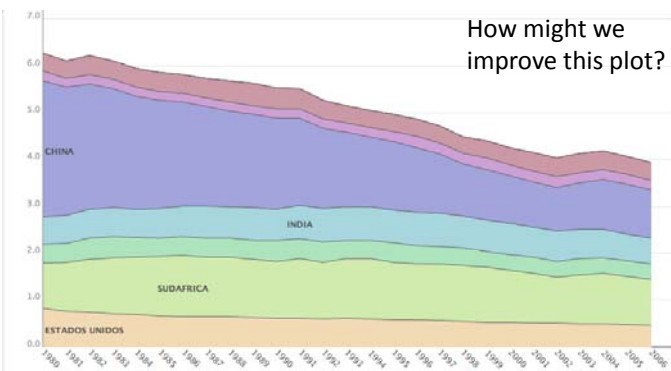


```
mycolors[1]
```

[1] "#A50026" - R doesn't give regular English names to these colors...

Case: CO₂ emissions around the world

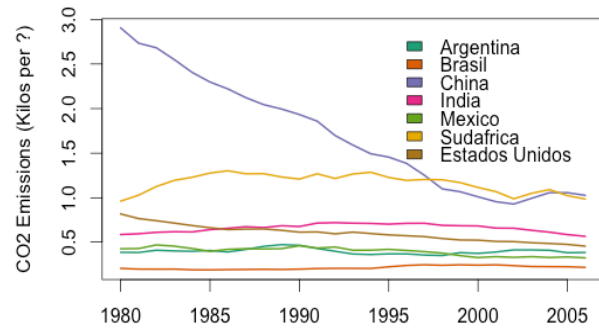
ManyEyes and CO₂



Changes

- Superpose rather than stack the curves so the baseline doesn't jiggle
- Use color on the lines rather than filling polygons

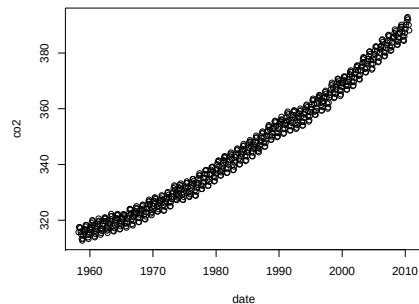
What can you see now?



Case: CO2 levels at Mauna Loa

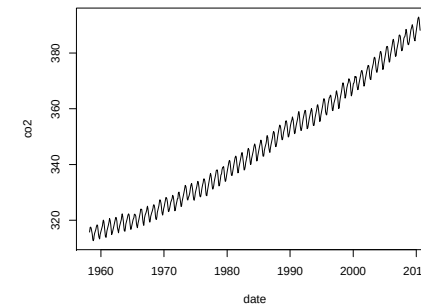
Time and the horizontal axis

Time Series – Pairs: (time, CO₂)

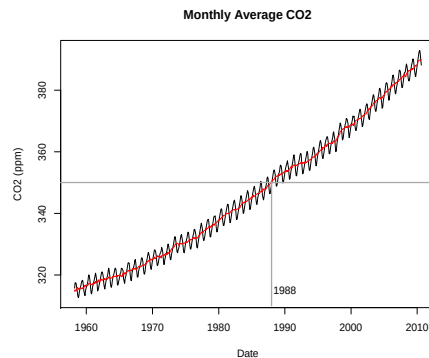


Points are typically not the best way to plot time series

Connect the measurements with line segments



Seasonality vs the long-term Trend



Aspect Ratio

- The height/width of the data region was selected to be about 1 so that the trend line is at about 45 degrees.
- The banking to 45 degrees let's us see that the curve is convex
- This means that the rate of increase of CO₂ is increasing through time

Including 0 & The Aspect Ratio

